GazeSpot Rebuttal (Submission 3106)

We thank the reviewers for the constructive comments and for recognizing the importance of crowded gaze target estimation, the simple frozen-VFM pipeline, and the motivation behind GGSF/SASA. GazeSpot's central contribution is to convert the standard head-box input into *head-conditioned spatial gating* before aggregation, then use *adaptive multi-layer routing* to balance geometry and semantics. The added OOD, noise, efficiency, control, and feature analyses below collectively reinforce this design goal.

GGSF as explicit head-conditioned focus.

Related reviewer concerns: fs7N-weakness-1; CoU9-weakness-2; 7pTL-weakness-1.

Our wording was intended to emphasize the induced spatial focus; the actual design contribution is the head-conditioned gate before feature aggregation. (1) It is *explicit* because GGSF computes a visible multiplicative mask M_{geo} and applies it to each VFM feature map before routing, rather than relying on implicit attention alone. (2) It is *head-conditioned* because this mask is generated from the target head's relative geometry, i.e., coordinates and scale $[x - c_x, y - c_y, w, h]$, so the resulting spatial focus is explicitly tied to the queried person.

Thus, (1) GGSF does not add separate head-pose, depth, object, or gaze-vector estimators; it uses the already-required head box to impose a structured focus that suppresses crowded distractors while preserving a simple frozen-VFM pipeline. Importantly, the final gaze direction is not determined by this mask alone, but inferred from the VFM image evidence. (2) The spotlight is not learned from mask labels: head-relative geometry constrains the gate, and heatmap/in-out losses optimize it end-to-end. (3) We will revise over-physical wording such as "physical frustum/gaze cone" to avoid misleading readers and use "frustum-inspired head-conditioned spatial gate" consistently.

Evidence beyond the Crowd subset.

Related reviewer concerns: uFer-weakness-1; CoU9-weakness-3.

To test whether the design only benefits our VAT Crowd subset, we add GOO-real under the same split, preprocessing, checkpoint-selection rule, and evaluation script as Reproduced Gaze-LLE. Table 1 shows that GazeSpot improves both AUC and L2, supporting generalization beyond the constructed subset. Single GT per sample makes Avg/Min L2 identical. We include GOO-real as a reproducible OOD test; ChildPlay will be evaluated once reliable access is available.

Table 1: GOO-real OOD evaluation.

Method	AUC↑	Avg L2↓	Min L2↓
Reproduced Gaze-LLE	0.8583	0.1997	0.1997
GazeSpot	**0.8833**	**0.1761**	**0.1761**

Controlled head-box noise analysis.

Related reviewer concerns: uFer-weakness-2; fs7N-weakness-2; epmB-weakness-1; 7pTL-weakness-5.

Head boxes are standard in gaze target estimation; GazeSpot turns this input into explicit spatial focus before VFM feature aggregation. We therefore quantify sensitivity with controlled translation/scale jitter instead of treating the input as perfect. Table 2 shows that under controlled jitter on the hardest VAT $>$ 4 split, GazeSpot remains stable and consistently outperforms Reproduced Gaze-LLE. Small non-monotonic variations come from stochastic jitter evaluation, and the revision will report this protocol.

Table 2: VAT $>$ 4 bbox-noise stress test.

Jitter	Method	AUC↑	L2↓	In/Out AP↑
0%	Reproduced Gaze-LLE	0.8751	0.1823	0.9449
0%	GazeSpot	**0.8898**	**0.1714**	**0.9671**
5%	Reproduced Gaze-LLE	0.8746	0.1837	0.9452
5%	GazeSpot	**0.8901**	**0.1707**	**0.9674**
10%	Reproduced Gaze-LLE	0.8739	0.1848	0.9450
10%	GazeSpot	**0.8902**	**0.1688**	**0.9673**
20%	Reproduced Gaze-LLE	0.8725	0.1856	0.9454
20%	GazeSpot	**0.8879**	**0.1699**	**0.9660**

Controls for spatial focus and adaptive routing.

Related reviewer concerns: CoU9-weakness-1; 7pTL-weakness-5.

Crowded gaze target estimation requires two linked controls. (1) For layer routing, Table 3 shows that raw concatenation and equal weighting keep routing static or generic and are clearly weaker than SASA on VAT Crowd ≥ 4. (2) Spatial-prior controls refine our claim: fixed Gaussian is much weaker than GGSF (AUC 0.8360 vs. 0.9260; L2 0.2122 vs. 0.1240; AP 0.9187 vs. 0.9261), while coordinate-conditioned controls show that the key factor is learnable head-relative spatial conditioning, not a hand-crafted physical mask. We will present the full table and avoid overstating GGSF as a physically exact frustum.

Table 3: SASA fusion controls on VAT Crowd ≥ 4.

Fusion	AUC↑	L2↓	AP↑
raw concat	0.8099	0.2358	0.9008
equal weight	0.9041	0.1405	0.9256
SASA	**0.9258**	**0.1239**	**0.9266**

Efficiency and parameter transparency.

Related reviewer concerns: epmB-weakness-2; CoU9-weakness-5.

GazeSpot uses a small trainable routing/gating overhead. Compared with the single-layer Reproduced Gaze-LLE baseline, the added cost is moderate: trainable parameters increase from 2.86M to 3.55M, and latency from 18.29 to 23.17 ms/img on VAT with RTX 3090 batch-1 evaluation, while the speed remains practical (43.17 FPS). This quantifies an acceptable accuracy–efficiency trade-off for improved crowded-scene robustness. We will report full parameter and latency details in the revision or supplementary material.

DINOv3 feature evidence.

Related reviewer concerns: 7pTL-weakness-2.

The feature claim is not that DINOv3 is generally worse, but that last-layer-only decoding is insufficient for crowded gaze target estimation. As stated in the paper, this phenomenon does not imply that DINOv3 is unsuitable; rather, GazeSpot uses SASA routing and GGSF spatial focus to unlock DINOv3 by combining early high-frequency geometry with deep semantics. Table 4 reports token cosine for crowd-token homogenization, plus boundary separability and FG/BG contrast for discriminability.

Table 4: VAT Crowd dense last-layer feature metrics.

Backbone	Token cosine↓	Boundary sep.↑	FG/BG contrast↑
DINOv2 last	**0.2850**	**0.3723**	**0.3823**
DINOv3 last	0.5342	0.2078	0.1673

Result scope and In/Out AP analysis.

Related reviewer concerns: 7pTL-weakness-3; fs7N-weakness-3.

(1) Regarding uneven gains across environments, GazeSpot is designed to suppress crowded-scene interference, so larger improvements are expected on dense/difficult subsets; sparse scenes contain fewer distractors and leave less room for improvement over strong baselines. We will clarify this scope instead of claiming uniform gains across all environments. (2) For In/Out AP, in the original paper's Table 2, GazeSpot-ViT-B improves VAT AUC/L2 but has lower In/Out AP, while GazeSpot-ViT-L reaches SOTA In/Out AP (0.912). We attribute this to capacity sensitivity: in/out prediction needs global evidence about whether the target leaves the frame, whereas GGSF/SASA most directly improve crowded spatial localization through head-conditioned focus and adaptive routing between geometry and semantics.

Crowd-subset reproducibility and final-paper precision.

Related reviewer concerns: 7pTL-weakness-4; CoU9-weakness-4, CoU9-weakness-5.

The Crowd subset is deterministically generated from VAT annotations by counting annotated head boxes per frame. To make the subset reproducible and auditable, the final version will release the construction script, frame/head IDs, bucket definitions, sample counts, evaluation code, and confound summaries such as in/out-frame ratio and bucket statistics. We will also fix equation punctuation/formatting and parameter-label precision.